CAPSTONE PROJECT | REVIEW 1 | REQUIREMENTS, LITERATURE SURVEY AND PROPOSED DESIGN

Project Title:Predictive Modelling for Heart Disease Risk Assessment Using Logistic Regression in Machine Learning

Team ID: G11 Batch: 2023–2027 Semester: 7 / 8

Project Guide: Dr. B. Perumal

Department of Electronics and Communication Engineering
Kalasalingam Academy of Research and Education | Date of Review: [14 october,2026]

Team Formation and Roles

Every member should be able to explain the whole project, not only their own module.

S.No	Name	Register No.	Relevant Skills / Prior Work	Role and Subsystem
1	*S. Yashika*	*9923005281*	*User Interface & Visualization Unit*	*User Interface & Visualization Unit (Streamlit Web Framework and Plotly real-time charts)*
2	*S. Abarna*	*9923005233*	*Processing & ML Intelligence Unit*	*Processing & ML Intelligence Unit (Python, Scikit-Learn, Logistic Regression model and StandardScaler)*
3	*Animala GuruSaiNath*	*9923005003*	*Sensor & Edge Acquisition Unit*	*Sensor & Edge Acquisition Unit (ESP32, MAX30102, AD8232 hardware wiring and serial telemetry)*

Project Guide: *Dr. B. Perumal*
Title approval by Reviewer Committee: *Approved*
Team repository: *[https://github.com/ece-kalasalingam/capstone-odd-team-G11] Created: Yes*

Need Analysis: Field Evidence

PI1 · CO1 · 20 marks
Team-level (with next slide)

Summarises Step 2 §1–2. Looks for: 2+ evidence types, quantified findings, proof attached.

Evidence Type	Source / Respondents	Date	Key Finding (with numbers)	Proof (Appendix #)
Survey, 10+ respondents	*public & university peers (13 respondents)*	*10.9.2026*	*76.9% lack real-time risk screening tools*	*Appendix 1*
government report	*Ministry of Health and Family Welfare (MoHFW) Report*	*15.9.2026*	*CVDs cause >28% deaths due to late diagnosis*	*Appendix 2*

Where and how we noticed the problem: *Gap identified via online survey & MoHFW report.*

Who is affected, and roughly how many: *At-risk CVD individuals lacking regular screening (>28% Indian deaths).*

Evidence note: *Primary survey data (13 users) verified via government stats.*

Root Cause, Impact and Problem Statement

PI1 · CO1 · 20 marks
Team-level (with previous slide)

Summarises Step 2 §3–6. Looks for: a true root cause and a solution-free statement.

ROOT CAUSE → FINAL PROBLEM

WHY 1 — Symptom: Delayed detection of cardiovascular risk during early non-emergency stages.

WHY 2 — Cause: Lack of continuous, real-time physiological health monitoring tools at home/clinics.

WHY 3 — Root cause: High cost and physical accessibility barriers of institutional diagnostic equipment.

FINAL PROBLEM STATEMENT

At-risk individuals face unmonitored cardiovascular anomalies, causing delayed diagnosis and high mortality. A successful solution must offer affordable, real-time risk screening.

[28%]

CVD share of total deaths in India due to delayed detection (MoHFW Report)

[76.9]

Survey Respondents Willing to Use the Proposed Heart Disease Risk Assessment System (Survey Data)

Finalised title: *Predictive Modelling for Heart Disease Risk Assessment Using Logistic Regression in Machine Learning*

Literature Reviewed

PI2 · CO1 · 15 marks
Team-level (3 slides)

Summarises Step 3 §1. Looks for: 6+ papers in your own words, with specific limitations.

Paper (IEEE format)	What It Did / Approach Used	Limitation (for our problem)
[1] Zulkiflee & Rusiman (2021)	*Binary Logistic Regression on 270 patient records.*	*Small dataset; lacks wider population*
[2] Uddin & Halder (2021)	*Ensemble ML approach via multilayer dynamic system.*	*Too complex for lightweight web deployment.*
[3] Qian et al. (2022)	*CVD prediction using physical exam indicators in cohort study.*	*Cohort-dependent; needs custom validation.*
[4] Srinivasan et al. (2023)	*Active-learning technique on UCI database.*	*Differs from standard Logistic Regression workflow. .*
[5] García-Ordás et al. (2023)	*Deep learning with feature augmentation for risk.*	*High compute resource demands.*
[6] Al-Alshaikh et al. (2024)	*Evaluated multiple machine learning algorithms.*	*Performance varies; requires custom testing.*

Pattern across the papers: *Existing works lack low-cost IoT physical sensors combined with real-time Logistic Regression web screening.*

Existing Products, Patents and Repositories

PI2 · CO1 · 15 marks
Team-level (3 slides)

Summarises Step 3 §2–3. Looks for: 2+ real products, 1+ patent or repo, limitations stated.

Products / Systems

Product / System	Made By	What It Does	Limitation
OMRON Complete BP Monitor + ECG	*OMRON Healthcare*	*Combines BP & ECG tracking*	*No Logistic Regression or custom IoT*
Philips Wearable Biosensor Technology	*Philips*	*Monitors HR, respiration & vitals*	*Lacks clinical parameter ML prediction*

Patents (or open-source repositories for software projects)

Patent Title and Number / Repo Link	What It Covers	How Our Project Differs
US Patent US12040093B2	*ECG diagnostic via neural network*	*Uses Logistic Regression + IoT Streamlit GUI*
US Patent US11826012B2	*Cloud-based remote patient monitoring framework for cardiovascular metrics.*	*We use a low-cost ESP32 sensor node with a lightweight Streamlit Logistic Regression GUI.*

Use or inspect the products hands-on where possible; state what you searched (Google Patents, Indian patents) and found.

Gap Analysis and Justification

PI2 · CO1 · 15 marks
Team-level (3 slides)

Summarises Step 3 §4–5. Looks for: a gap that follows from the limitations just shown.

What Our Solution Needs to Do	Covered by Existing Work?	Which Refs / Products
Predict heart disease risk using clinical parameters	*Yes*	*Research Papers [1], [3], [4], [6]*
Provide a simple, understandable prediction interface	*Partly*	*Research Papers [2], [5]; Streamlit-based proposed interface*
Combine machine learning with IoT-based physiological monitoring	*Partly*	*OMRON BP Monitor + ECG Philips Wearable Biosensor*
Present preliminary results while encouraging professional medical assessment	*Partly*	*Research Papers [1]–[6]*

Justification for the project: *Existing research uses machine learning to predict heart disease risk, while commercial devices monitor physiological parameters. Our project integrates Logistic Regression, ESP32-based sensors (MAX30102 and AD8232), and a Streamlit interface for preliminary risk assessment. The system supports early screening but does not replace professional medical diagnosis.*

Functional and Performance Requirements

PI3 · CO1 · 15 marks
Team-level

Summarises Step 4 §1–2. Looks for: measurable, testable, traceable. Targets are baselined now.

Functional (5–8 in Step 4)

ID	The System Must Be Able To...	Need (Step 2)
FR1	*Accepts 13 clinical input attributes for heart disease risk prediction.*	*N1*
FR2	*Predicts heart disease risk using the trained Logistic Regression model*	*N1*
FR3	*Displays risk probability and risk category in the Streamlit interface.*	*N1*
FR4	*Collects heart rate,SpO2,and ECG signals using ESP32-connected sensors*	*N2*
FR5	*Shows sensor readings and predicted heart disease risk together in Streamlit.*	*N2*

Performance (3–5 in Step 4)

ID	Measured	Target	Verified By	Links (optional)
PR1	*Accuracy*	*86.89% baseline*	*Model evaluation*	*FR2*
PR2	*ROC-AUC*	*95.35% baseline*	*ROC-AUC evaluation*	*FR2*
PR3	*Input attributes*	*13 attributes*	*Input validation*	*FR1*
PR4	*Sensor readings*	*Heart rate, SpO_2 and ECG*	*Sensor testing*	*FR4*

Use the exact IDs from Step 4 on every later slide. Vague wording such as "user-friendly" or "works well" scores Level 0. Targets are baselined at this review; Review 2 tests against them, and any later change must be documented, justified and approved by the guide.

Standards, Constraints and SDGs

PI4 · CO2 · 10 marks
Team-level

Summarises Step 4 §3–6. Looks for: cited standards, every category addressed, SDG link specific.

Constraint Type	Details, or Why Not Applicable
Economic	*Low-cost components reduce overall project cost.*
Environmental / Sustainability	*Low-power components reduce energy consumption.*
Manufacturability / Feasibility	*ESP32 and sensors enable easy prototyping.*
Ethical / Health and Safety	*The system supports screening, not diagnosis.*
Social	*Provides an accessible interface for users.*
Regulatory / Legal / Political	*Medical and data privacy requirements must be considered.*
Data ethics and management	*Patient data must be kept private.*

Standards / Codes

- *IEEE 802.11: ESP32 Wi-Fi communication.*
- *ISO/IEC 25010:2023: Software quality evaluation.*

SDGs (1–3)

- *SDG 3 Good Health and Well-being: Supports early heart disease risk screening*
- *Target :Improve early risk awareness*

Objectives and Scope

Objectives (specific and measurable)

1. *O1: Develop a Logistic Regression model using 13 clinical attributes*
2. *O2: Display heart disease risk prediction through a Streamlit web application*
3. *O3:Integrate ESP32,MAX30102, and AD8232 for Physiological Monitoring*

Candidate Design Approaches

PI5 · CO2 · 15 marks
Team-level (1 slide)

Summarises Step 5 §1. Looks for: 2–3 approaches that differ in core method, not components.

Approach	Core Method and Brief Description	Key Advantage	Key Drawback
A	*Logistic Regression using clinical data*	*Simple and fast*	*Manual inputs required*
B	*Machine Learning with IoT sensors*	*Combines prediction and monitoring*	*Sensor integration needed*
C	*Deep Learning for heart disease prediction*	*Learns complex patterns*	*Requires more data and computing*

Check: *[Would a reviewer call these three different ideas, or one idea with different parts swapped in? Only list approaches the team seriously considered.]*

Comparison and Final Selection

PI5 · CO2 · 15 marks
Team-level (1 slide)

Summarises Step 5 §2–4. Looks for: criteria from your requirements, evidence-based choice.

Criterion (3–5)	Weight	A: Score + Note	B: Score + Note	C: Score + Note
Accuracy potential	*0.30*	*3 - Baseline*	*4-Combined system*	*4 – Complex patterns*
Cost	*0.25*	*4 – Low cost*	*3 – Sensor cost*	*2 – Higher resources*
Ease of implementation	*0.25*	*4 - Simple*	*3 – Integration needed*	*2 – Complex*
Physiological monitoring	*0.20*	*1 – Not included*	*4 – Sensors included*	*1 – Not included*
Weighted score	**1.00**	**3.10**	**3.55**	**2.55**

Use this weighted table (Option 2 in Step 5), or show the qualitative comparison (Option 1) if that is what you submitted. Weights add to 1.00; scores run 1–4, with 4 best.

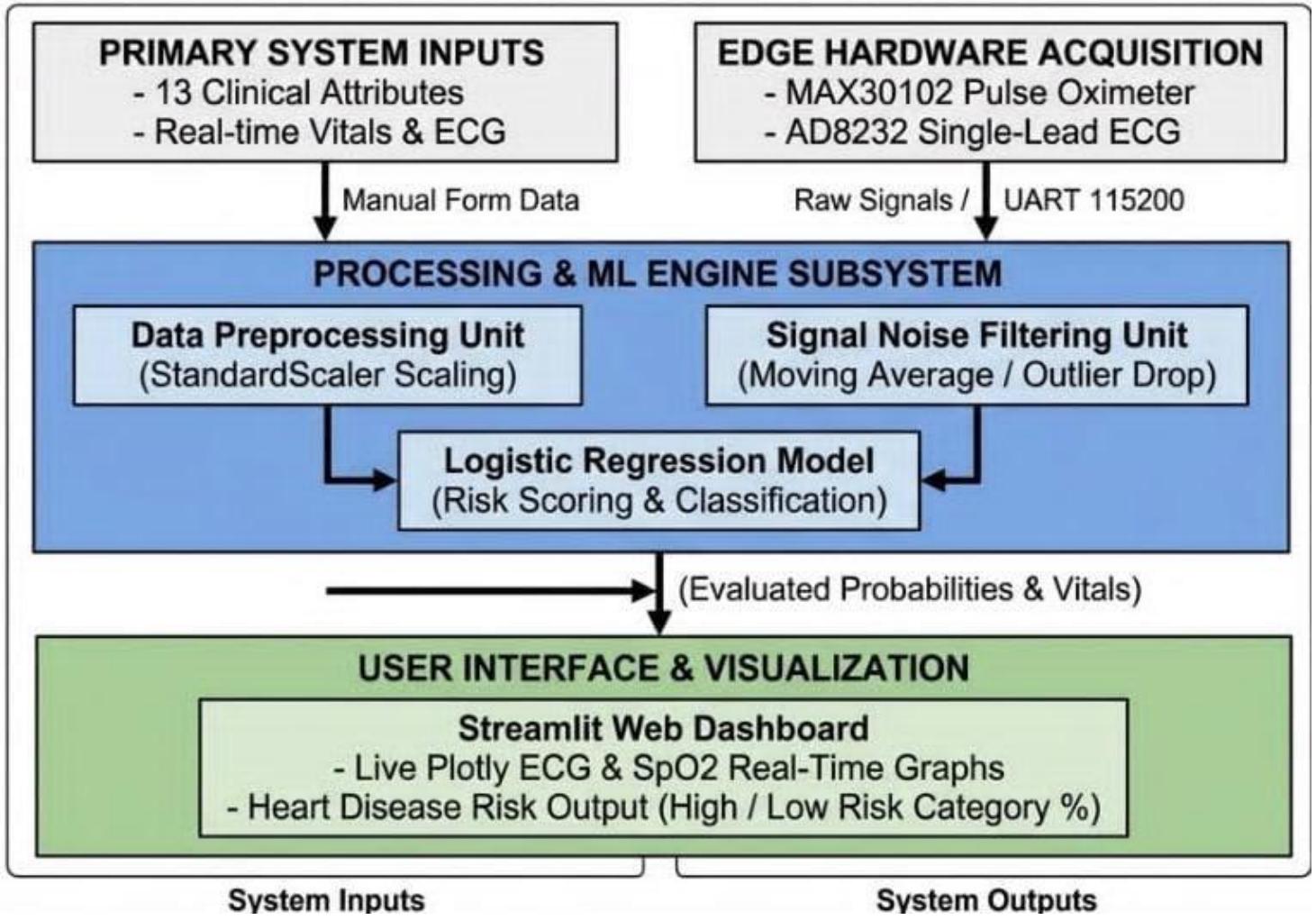
Selected approach and why: *Approach B – ML + IoT, combining Logistic Regression prediction with physiological monitoring.*

Novelty vs Step 3, IP overlap, TRL (1–9): *Integrated prediction and monitoring. IP overlap: Existing technologies. TRL: 3 (prototype validation stage).*

System Architecture and Traceability

PI6 · CO2 · 10 marks
Team-level (1 slide)

Summarises Step 6 §1–2. Looks for: legible diagram, every requirement mapped. Proposed design only.



System inputs: *[13 Clinical Attributes, Sensor Vitals (Heart Rate, SpO2, ECG)]*

System outputs: *[Heart Disease Risk Probability (%), Real-time Streamlit Dashboard Display]*

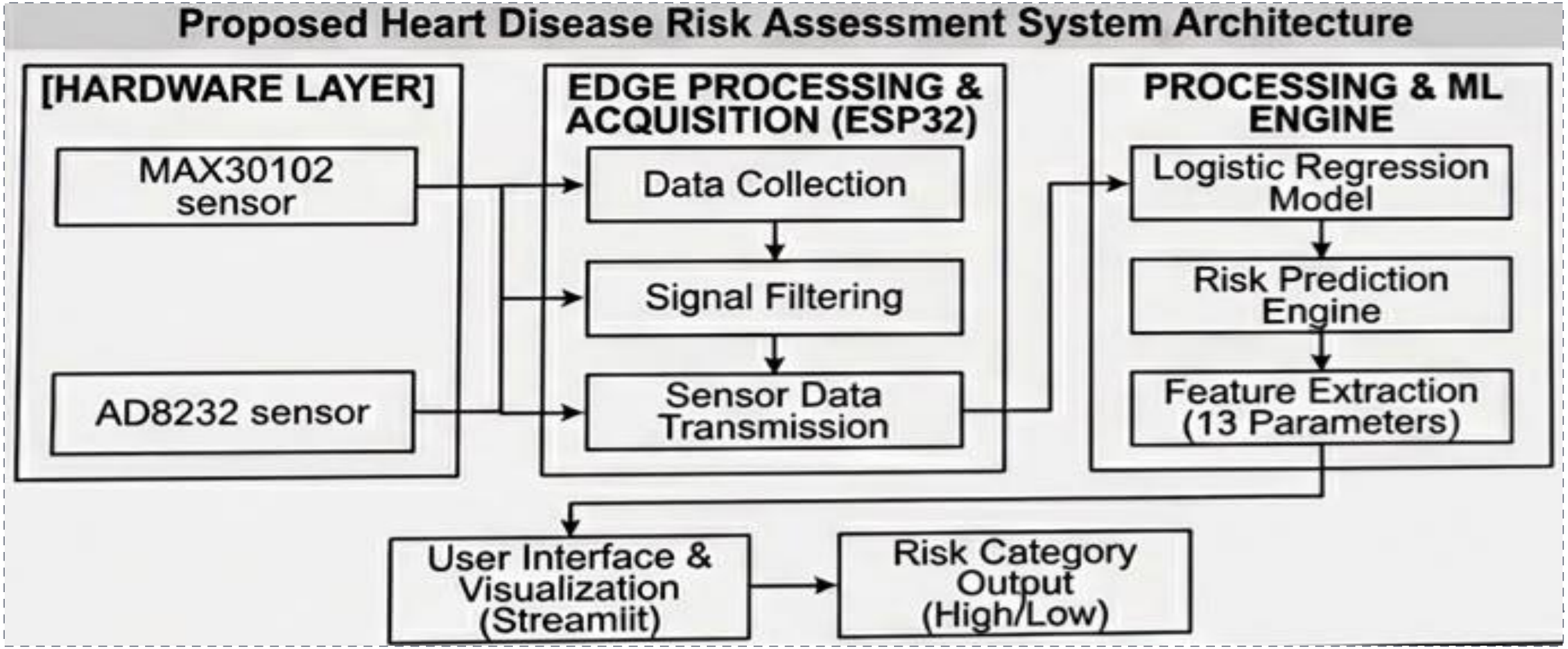
Req. ID	Requirement	Subsystem
FR1	*Accept 13 clinical inputs*	*S1*
FR2	*Predict heart disease risk*	*S2*
FR3	*Risk output*	*S2*
PR1	*86.89% accuracy*	*S3*

Every functional requirement needs a subsystem; every subsystem here needs a block in the diagram.

System Architecture and Traceability

PI6 · CO2 · 10 marks
Team-level (1 slide)

Summarises Step 6 §1–2. Looks for: legible diagram, every requirement mapped. Proposed design only.



Work Breakdown and Responsibilities

PI7 · CO5 · 15 marks
Individual-level (6 slides)

Summarises Step 1 §1–2. Looks for: a named owner for every subtask. No 'all team members'.

Task	Subtask Owner (Reg. No.)
1.0 Literature Review	*1.1 [9923005281] · 1.2 [9923005233] · 1.3 [9923005003] · 1.4 [9923005281] · 1.5 [9923005233]*
2.0 Requirements and Constraints	*2.1 [9923005233] · 2.2 [9923005281] · 2.3 [9923005003] · 2.4 [9923005003]*
3.0 Concept Development	*3.1 [9923005281] · 3.2 [9923005233] · 3.3 [9923005003]*
4.0 Final Design and Implementation	*4.1 [9923005003] · 4.2 [9923005281] · 4.3 [9923005233] · 4.4 [9923005003]*
5.0 Testing and Evaluation	*5.1 [9923005281] · 5.2 [9923005233] · 5.3 [9923005003] · 5.4 [Reg]*
6.0 Report and Final Presentation	*6.1 [9923005233] · 6.2 [9923005281] · 6.3 [9923005003]*

Per-Student View

- *Student A [9923005233]: subtasks [1.1,1.4,2.1,2.4,3.1,4.2,5.1,6.1]; deliverables [Report, hardware plan, test results]*
- *Student B [9923005281]: subtasks [1.2,1.5,2.2,3.3,4.1,4.4,5.3,6.2]; deliverables [Report, hardware plan, test results]*
- *Student C [9923005003.]: deliverables [Report, hardware plan, test results]*

Every subtask has a named owner. Tasks 4.0 and 5.0 are expected to loop: testing can send the design back for revision (subtask 5.4).

Individual Contributions

PI7 · CO5 · 15 marks
Individual-level (6 slides)

Each member presents their individual contribution to the completed pre-Review 1 activities, supported by verifiable documentary evidence.

Member	Work Done This Period	Evidence (commits, log, report section)
[Abarna S]	*[Dataset preprocessing and Logistic Regression model development]*	*[Python code, model results]*
[Yashika S]	*[Streamlit interface development and prediction display]*	*[UI screenshots, source code]*
[Animala guru sainath]	*[ESP32 sensor integration, testing and report documentation]*	*[Hardware photos, report pages]*

Timeline

PI7 · CO5 · 15 marks
Individual-level (6 slides)

Summarises Step 1 §3. Shift the marked weeks to match your actual duration and pace.

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
1.0 Literature Review	W1	W2	W3									
2.0 Requirements		W2	W3	W4								
3.0 Concept Development				W4	W5							
4.0 Design / Implementation					W5	W6	W7	W8	W9	W10		
5.0 Testing								W8	W9	W10	W11	
6.0 Reporting										W10	W11	

Key milestones: *[Subsystem prototypes by Wk _; integration by Wk _; testing complete by Wk _. Pale bars are only the Step 1 suggested pattern, not an approved schedule: replace them with your actual plan.]*

Budget Estimation

PI7 · CO5 · 15 marks
Individual-level (6 slides)

Summarises Budget Estimation §1–4. Looks for: BoM linked to subsystems, evidence of prices.

Hardware / Components (preliminary BoM)

Subsystem	Component	Specification / Model	Qty	Unit ₹	Total ₹
[1]	*Development Board*	*ESP32*	*1*	*450*	*450*
[2]	*ECG & Heart Rate Sensors*	*AD8232+MAX30102*	*2*	*1130*	*1130*
[3]	*OLED Display*	*OLED Module*	*1*	*180*	*180*
[4]	*Breadboard, PCB, Wires, LEDs & Buzzer*	*Basic components & USB Cables*	*1set*	*395*	*395*

Total Estimated Cost

Cost Head	₹
Hardware	*[2155]*
Software / online	*[0]*
Other expenses	*[108]*
Total	**[₹ 2263]**

Key component selection: *[ESP32 (Wi-Fi/IoT); AD8232 & MAX30102 (ECG/SPO2)]*

Evidence attached: *[Datasheets & Receipts]*

Software and other expenses: *[Open-source (₹0); Printing (₹108)]*

Note: *Do not invent quotations; university-owned resources are ₹0, not purchases.*

Risk Register

PI7 · CO5 · 15 marks
Individual-level (6 slides)

Each member states the risks to their own subtasks. Looks for: specific, credible risks with a mitigation plan.

Risk	L / I	Mitigation
[Sensor Data Noise] · Owner [9923005281]	*M/H*	*Digital software filtering*
[Component Delivery Delay] · Owner [9923005233]	*M/M*	*Local vendor procurement*
[ML Model Overfitting] · Owner [9923005003]	*L/H*	*Dataset cross-validation*
[Wi-Fi Connectivity Lag] · Owner [9923005281]	*M/M*	*Offline data buffering*

Each member shall explain the risks associated with their assigned subtasks, the potential impact on project execution, and the proposed mitigation. Team-level risks are permitted, provided that each risk is specific, justified, and assigned to a responsible member.

Next Review Commitments

PI7 · CO5 · 15 marks
Individual-level (6 slides)

Looks for: measurable commitments with a named owner, due week and verifiable evidence.

Commitments for Review 2

Deliverable	Owner (Reg. No.)	Due	Evidence to Verify
Complete sensor integration	*[9923005233]*	*[Week 6]*	*Sensor test photos*
Test Logistic Regression model	*[9923005281]*	*[Week 5]*	*Accuracy results*
Integrate Streamlit interface	*[9923005003]*	*[Week 7]*	*Interface screenshots*
Test complete system	*[9923005233]*	*[Week 8]*	*Test results*